

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in larger structures. Their significance depends on whether their galaxies differ from comparable regular-group members, not only from field galaxies.

Aims. We test whether galaxies in compact groups of four members (CG₄) remain special against regular groups of four members (RG₄), the four brightest galaxies of richer regular groups (Control_{4B}), and projected compact cores formed by the brightest group galaxy and its three nearest projected companions (Control_{4C}).

Methods. From previous samples, we retain 62 non-split CG₄ systems. We combine Galaxy Zoo 1 vote fractions, Sloan Digital Sky Survey stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, and seeing-corrected half-light radii. Inference uses per-control adjusted galaxy-level models, galaxy-level matching, group-level satellite-composition matching, and within-host stratified comparisons.

Results. The main residual signal is morphological and graded by control construction. CG₄ galaxies, especially satellites, have higher elliptical-vote and lower spiral-vote fractions than regular-group analogues. The adjusted individual-galaxy excess is strongest against Control_{4B} and RG₄ and smallest against Control_{4C}. The matched satellite-composition contrast is positive against all controls, clearest against Control_{4B} and RG₄ and weakest against Control_{4C}, with the same hierarchy under satellite-mass balancing. No additional within-host excess is detected after conditioning on host-centric radius. The morphology result is stable to vote-threshold, vote-fraction, Sérsic-index, and close-pair checks, and is attenuated by a projected pairwise tidal-density proxy. Star-formation, colour, active-galactic-nucleus, magnitude-gap, H α , and fixed-mass size diagnostics show no comparably robust CG₄ offsets.

Conclusions. CG₄ systems therefore host a population shifted towards early-type-like morphologies, consistent with accumulated transformation in dense group cores and/or preprocessing, without establishing a present-day quenching mechanism unique to compact groups.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson 1982; Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field (building on the photometric morphologies of Hickson et al. 1989 and the merger-remnant expectations of Zepf & Whitmore 1993), in line with the broader morphology–density relation (Dressler 1980); multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and stellar-population properties relative to looser environments (Johnson et al. 2007; Tzanavaris et al. 2010;

Bitsakis et al. 2011; Coenda et al. 2012), and their H I deficiency has been organised into an evolutionary sequence (Verdes-Montenegro et al. 2001).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies point to transformation in progress in compact groups, with stronger signatures in non-isolated systems and host structures (Montaguth et al. 2023, 2025, 2026a,b), but they do not by themselves answer this control question.

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In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the cores of richer groups. Here we ask the complementary question: even if compact groups are not strongly distinct as systems, are their member galaxies individually different? We compare CG₄ member galaxies with three regular-group analogues – RG₄, Control_{4B}, and Control_{4C} (Sect. 2.1) – controlling for stellar mass, redshift, galaxy rank, and available group-scale properties, and use complementary individual-galaxy, group-composition, and shared-host contrasts to identify which observable, if any, carries a residual compact-group signal.

Section 2 describes the samples, the classifications, and the statistical design. Section 3 presents the population contrasts, the matched and within-host comparisons, and the robustness and secondary diagnostics. Section 4 discusses the physical interpretation, and Section 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025), whose construction we briefly recall. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume- and luminosity-complete limits $0.005 \leq z_{\text{BGG}} \leq 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all four galaxies lying within 3 magnitudes of the brightest group galaxy (BGG). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups.

The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. The three controls serve different physical purposes. RG₄ contains regular groups with exactly four members (excluding systems identical to a CG₄) and is the most direct equal-membership comparison; it is also by far the loosest: the median (over groups) of the median pairwise projected member separation is 430 kpc for RG₄ (interquartile range 292–494 kpc, proper, angular-diameter distances) against 143 kpc for CG₄ (114–177 kpc), so RG₄ quartets are typically ~3 times more extended than the compact groups they are compared with. Control_{4B} contains the four brightest galaxies of each richer regular group and tests whether the CG₄ population resembles the luminous core of an ordinary group; its corresponding median pairwise separation is 447 kpc. Control_{4C} contains the BGG plus its three closest projected companions, selected among members within 3 magnitudes of the BGG, and is the most compact-core-like ordinary-group analogue: its median pairwise separation

is 293 kpc (IQR 187–436 kpc), intermediate between CG₄ and the looser controls. The CG₄–Control_{4C} overlap coefficient in the compactness score $-\log R_{\text{pair,med}}$ is 0.34, and 18% of CG₄ groups lie above the Control_{4C} 95th percentile in compactness, so Control_{4C} provides common support but is not a compactness-matched sample; it is regenerated for this paper directly from the 765 parent groups¹. Every control group containing at least one CG₄ galaxy is excluded from the control samples, so no galaxy appears on both sides of any comparison; the extent of this compact-group/projected-core overlap is quantified in Appendix C. We also remove the systems associated with Lim group 3688, which appears to be spurious: its single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) is selected into both our Control_{4B} and Control_{4C} samples and inflates the group velocity dispersions to $\approx 2000 \text{ km s}^{-1}$ (1979 and 2042 km s^{-1} , respectively). After these removals, the control samples contain 698 Control_{4B}, 703 Control_{4C}, and 56 RG₄ groups (2792, 2812, and 224 galaxies); the Control_{4B} and Control_{4C} counts differ because the luminosity-ranked and projected-distance-ranked core definitions select different galaxies and therefore retain different Lim groups after the compact-group exclusion.²

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); the exact columns, table joins, and query are given in Appendix A. Throughout, ‘SDSS selection’ denotes the reference sample returned by that query: all spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and Galaxy Zoo classifications. It serves as a field-dominated reference population, not as a control sample. All projected distances and derived quantities adopt the Planck 2015 cosmology (Planck Collaboration et al. 2016), consistently with our previous paper.

¹ The $\Delta m \leq 3$ eligibility filter is applied before the projected-distance ranking, as in the construction behind our previous paper’s published statistics, whose counts (61 excluded projected-core groups, 704 retained) this regeneration reproduces exactly. Machine-verified construction assertions (every companion within 3 magnitudes; every quartet the three nearest such members) are provided with the analysis repository (`referee/TO_control4c_audit.py`).

² Two unit conventions inherited from the parent pipeline are documented for reproducibility. (i) The exported per-galaxy projected BGG separations (`dist2BGG`) are great-circle angles in radians multiplied by 3600 (i.e. 4.7% larger than arcminutes); every use in this paper divides by 3600 before converting with the angular-diameter distance, and all rankings are unaffected. (ii) The inherited group size (and hence the crossing time) converts angular size with the *luminosity* distance, overestimating proper sizes by $(1+z)^2$ (3.3–9.6%, median 7.2% over the sample), whereas the virial mass uses the angular-diameter distance; we keep the inherited convention for cross-paper comparability with our previous paper’s published sizes, no fitted model uses size or crossing time as a covariate, and the normalized host-centric radii of the radial diagnostics inherit only this smooth $(1+z)^2$ scale factor, which leaves ranked and z -adjusted diagnostics unchanged.

2.2. Star-formation and morphology classes

We classify galaxies from SDSS total sSFRs. In the MPA-JHU catalogue, values of `-9999` in the sSFR fields are sentinel values indicating that no valid sSFR estimate is available, rather than physical measurements; the same missing-value convention is used for MPA-JHU DR8 target quantities by Zeraatgari et al. (2024). We therefore treat `specsfr_tot_p50 = -9999` as *missing data*: such galaxies are excluded from the star-formation classification and from every sSFR-based figure and fraction, and are reported as counts only (Table 1). All quenched and star-forming fractions quoted in this paper are computed among classified galaxies only. The missing-sSFR incidence is markedly higher in CG₄ (7% of members, and 10% of BGGs) than in the field selection (1%), plausibly reflecting crowded-field spectroscopy and deblending (Sect. 4.3) rather than a physical population. Galaxies with a measured sSFR are split into quenched and star-forming classes with a two-component Gaussian mixture model (GMM) in the $\log M_\star$ – $\log$ sSFR plane; BPT/AGN handling and the exact GMM boundary definition are given in Appendix B (Fig. B.1).

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), a single-question classification in which volunteers assigned each galaxy image to one of several morphological categories. The resulting vote fractions, debiased for the redshift-dependent classification bias following Bamford et al. (2009), are provided in the `zooSpec` table through the columns `p_el_debiased` (fraction voting “elliptical”) and `p_cs_debiased` (fraction voting combined-spiral; Lintott et al. 2011). We classify a galaxy as elliptical if `p_el_debiased` exceeds 0.5, as spiral if `p_cs_debiased` exceeds 0.5, and as “uncertain” otherwise. These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo 1 elliptical-vote proxies for early-type morphology rather than as a bulge-disc structural classification.

2.3. Galaxy sizes

Our primary galaxy-size measure is the circular half-light radius of the pure Sérsic fit in the r band, $R_{\text{chl},r}$, from the bulge+disk decomposition catalogue of Simard et al. (2011). These are PSF-convolved GIM2D fits with re-derived sky levels and deblending designed with crowded fields in mind – the relevant regime for compact-group photometry – so the fitted sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer `SpecDR7` cross-match table. Two guards protect this bridge: rows in which the Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and their structural values dropped (9 rows), and rows in which two distinct catalogue galaxies of the same group resolve to a single DR7 identifier are treated as blended detections and dropped (0 rows). The Simard sizes are tabulated in kpc under the catalogue’s own cosmology; we convert them back to angular sizes with the catalogue plate scale and then to proper kpc with the same Planck 2015 conversion as every other projected quantity in this paper.

As a secondary size measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$. Petrosian radii are not corrected for seeing, and the SDSS Petrosian aperture misses a profile-dependent fraction of the flux, roughly 20% for an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005) – precisely the elliptical/early-type-like population in which CG₄ is enhanced. The Petrosian measure is nevertheless retained because it is model-independent, essentially fully available, and provides the like-for-like bridge to the compact-group size study of Coenda et al. (2012). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 60 galaxies) are excluded from the primary sample — per sample, 5 of 197 Simard-matched CG₄ members (2.5%), against 22/1939, 31/2061, and 2/199 in Control_{4B}, Control_{4C}, and RG₄ (1.0–1.5%): the cut is mildly differential in the direction expected if pegged fits preferentially affect concentrated early types, so removing them is conservative with respect to the CG₄ early-type excess — and Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures, and its difference between CG₄ and the controls, is audited in Sect. 3.8.

2.4. Statistical design

Descriptive baseline comparisons use Fisher or Barnard exact tests for two-sample categorical tables, and rank-sum or bootstrap comparisons for distributional and median differences. Spearman and Pearson coefficients are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small.

The primary inference combines four complementary, non-hierarchical estimands: (i) per-control covariate-adjusted galaxy-level associations; (ii) one-to-one matched individual-galaxy contrasts; (iii) matched group-level differences in satellite composition; and (iv) within-host comparisons of CG₄ members with non-CG₄ co-members of the same host group. These analyses differ in statistical unit, conditioning set, outcome, effect scale, and interpretation.

Throughout, the independence unit is the *physical* group: standard errors are clustered by Lim group (unifying the overlapping control definitions, with CG₄ systems assigned to their host Lim group), and all resampling is blocked at the group level, since the four galaxies of one group are not independent observations. Pooled models across the three controls are reported only as secondary summaries, after deduplicating the overlapping control samples to one row per physical galaxy (Appendix F). Bootstrap p-values use the add-one empirical convention $p = (k+1)/(B+1)$ with the number of resamples B reported, so no resampling p-value can fall below its Monte-Carlo resolution; displayed p-values are floored at 10^{-4} .

Throughout, the velocity dispersion σ_v that enters models and matching is the quartet-level gapper estimate (Wainer & Thissen 1976) on the four selected members’ redshifts, converted as $c\sigma_z/(1+\bar{z})$ — for *all* samples, including Control_{4B} and Control_{4C} quartets embedded in richer hosts, whose σ_v therefore describes the selected quartet rather than the host halo (this is how the spurious Lim 3688

quartets of Sect. 2.1 reach $\approx 2000 \text{ km s}^{-1}$). With four tracers this estimator is intrinsically noisy: in a Gaussian Monte Carlo of 10^5 quartets, $\sigma_{\text{gapper}}/\sigma_{\text{true}}$ has median 0.96 with a 16–84% range of 0.57–1.42. Candidate covariates enter an adjusted model when non-missing for at least 65% of that model’s frame (matching variables require 70%); in practice σ_v is complete for every quartet and always enters, the group-luminosity, stellar-mass, redshift, and rank covariates are likewise complete, and no harmonised group-mass covariate is available (Appendix F.4). Because σ_v is noisy, every family that uses it — the per-control adjusted families, the pooled models, the galaxy-level propensity model, and the group-level matching — is also refit without it as a labelled sensitivity; no Holm-significant conclusion changes (Sects. 3.2, 3.3).

Holm or Benjamini–Hochberg corrections are applied within pre-defined families of related tests. These families are defined over the full set of computed diagnostics – including those reported only in the appendices – and are *not* recomputed over the subset of results shown in the main text. For the per-control group-level satellite-composition matches, the three control definitions are treated as separate scientific estimands because Control_{4B}, Control_{4C}, and RG₄ answer different control questions; their p-values are therefore not adjusted across control definitions, and the pooled match is a secondary summary rather than a fourth confirmatory comparison. The paired sign-flip permutation p-value is the primary paired test for these group-level matches; the bootstrap sign-crossing p-value and interval are used for robustness and effect-size uncertainty. A Holm sensitivity calculation for the three per-control bootstrap and permutation p-value sets is reported in Appendix F. Colours, AGN-like fractions, H α equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index are secondary diagnostics: the conclusions below rely on signals that survive the adjusted, matched, and within-host analyses, not on isolated exploratory tests.

3. Results

3.1. Raw population contrasts

The sSFR status and Galaxy Zoo morphology mix of each sample are shown in Tables 1 and 2. For the morphologies, Barnard’s two-sided exact tests give p-values of $p = 1.4 \times 10^{-3}$ for ellipticals and $p = 1.2 \times 10^{-3}$ for spirals versus the Control_{4B} sample, $p = 1.6 \times 10^{-3}$ and $p = 6 \times 10^{-3}$ versus Control_{4C}, and $p < 10^{-4}$ and $p < 10^{-4}$ versus RG₄. The galaxies in CG₄ systems, which we identified as frequent cores of richer groups in Tricottet et al. (2025), show a clear raw excess of elliptical classifications and a deficit of spiral classifications. Restricted to BGGs, CG₄ BGGs have a higher elliptical fraction (58.1%) than RG₄ BGGs (37.5%); the p-values obtained with two-sided Fisher exact test are 1.00 against Control_{4B}, 1.00 against Control_{4C} and 0.03 against RG₄. Restricted instead to star-forming galaxies, the elliptical/spiral mix of CG₄ is statistically consistent with that of every control (Barnard two-sided exact test: $p = 0.283$ against Control_{4B}, $p = 0.531$ against Control_{4C}, and $p = 0.186$ against RG₄); we detect neither an excess nor a deficit of *star-forming* elliptical galaxies, so the raw morphology excess is not carried by star-forming ellipticals.

Table 1. Number of galaxies in each sSFR class for each sample. Galaxies without a valid sSFR estimate (‘No sSFR’) are excluded from the classification: their percentages are quoted with respect to all galaxies of the (sub)sample, while the quenched and star-forming percentages are quoted with respect to classified galaxies only. The SDSS-selection row is a field-dominated reference reaching ~ 3 mag fainter than any CG₄ member, shown for orientation only; it is not a control sample and enters no test in this paper.

Sample	No sSFR	Quenched	Star-forming
CG ₄	18 (7%)	146 (63%)	84 (37%)
BGG	6 (10%)	45 (80%)	11 (20%)
Satellites	12 (6%)	101 (58%)	73 (42%)
Control _{4B}	180 (6%)	1593 (61%)	1019 (39%)
BGG	58 (8%)	533 (83%)	107 (17%)
Satellites	122 (6%)	1060 (54%)	912 (46%)
Control _{4C}	191 (7%)	1560 (60%)	1061 (40%)
BGG	59 (8%)	537 (83%)	107 (17%)
Satellites	132 (6%)	1023 (52%)	954 (48%)
RG ₄	5 (2%)	101 (46%)	118 (54%)
BGG	3 (5%)	38 (72%)	15 (28%)
Satellites	2 (1%)	63 (38%)	103 (62%)
SDSS selection	634 (1%)	7705 (15%)	44192 (85%)

Table 2. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample. As in Table 1, the SDSS-selection row is a field-dominated reference reaching ~ 3 mag fainter than any CG₄ member, not a control sample.

Sample	Elliptical	Spiral	Uncertain
CG ₄	124 (50%)	86 (35%)	38 (15%)
Control _{4B}	1106 (40%)	1266 (45%)	420 (15%)
Control _{4C}	1118 (40%)	1229 (44%)	465 (16%)
RG ₄	60 (27%)	129 (58%)	35 (16%)
SDSS selection	10091 (19%)	34715 (66%)	7725 (15%)

The star-formation contrast is weaker and rank-dependent. In the raw all-galaxy comparison, Fisher’s exact test gives star-forming-fraction p-values of $p = 0.481$ versus the Control_{4B} sample, $p = 0.262$ versus Control_{4C}, and $p = 3 \times 10^{-4}$ versus RG₄. For satellites, the corresponding p-values obtained with two-sided Fisher exact test are $p = 0.302$ against Control_{4B}, $p = 0.114$ against Control_{4C} and $p = 2.3 \times 10^{-4}$ against RG₄; for BGGs, they are $p = 0.578$, $p = 0.577$, and $p = 0.37$ (same control order). The satellite deficit is clearest against RG₄. We treat all raw tests as descriptive, because star-formation class and morphology are entangled with stellar mass and luminosity; this motivates the adjusted analyses of Sect. 3.2.

For the star-forming subset, we fit the star-forming main sequence with polynomials of order 1–4, adopting order 2 as the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.305 dex; Appendix D). The median CG₄-minus-control residual offsets about this relation are small, control-dependent, and change in strength after matching (Appendix D), so we treat main-sequence residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset.

3.2. Per-control adjusted contrasts (primary)

The three control samples answer different questions, so our primary galaxy-level inference fits CG₄ against each control

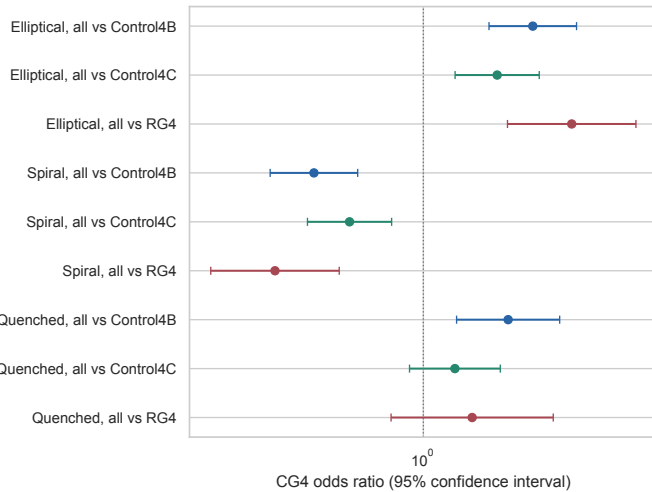


Fig. 1. Adjusted CG_4 odds ratios from the three separate per-control contrasts (colour-coded by control sample) for the all-member elliptical, spiral and quenched outcomes. Bars show 95% confidence intervals; the vertical line marks equal odds.

separately: against $Control_{4B}$ (is the CG_4 population special compared with the luminous population of richer ordinary groups?), against $Control_{4C}$ (compared with BGG-centred projected cores?), and against RG_4 (compared with true four-member ordinary groups?). Each contrast fits the same family of adjusted binomial models (stellar mass, redshift, rank, group luminosity, and velocity dispersion; the completeness rule of Sect. 2.4 retains all five for every contrast), with standard errors clustered by *physical* group: the Lim group for controls, and for CG_4 systems their host Lim group where one exists, so that the same physical system never enters as more than one cluster. Holm correction is applied within each contrast across its model family.

Against $Control_{4B}$, the elliptical odds ratio is 2.14 (95% CI [1.58, 2.90]; Holm-corrected $p < 10^{-4}$), and the quenched odds ratio is 1.80 (Holm-corrected $p = 0.003$).

Against $Control_{4C}$, the elliptical odds ratio is 1.67 (95% CI [1.25, 2.24]; Holm-corrected $p = 0.004$), and the quenched odds ratio is 1.25 (Holm-corrected $p = 0.346$).

Against RG_4 , the elliptical odds ratio is 2.81 (95% CI [1.79, 4.38]; Holm-corrected $p < 10^{-4}$), and the quenched odds ratio is 1.40 (Holm-corrected $p = 0.476$).

The gradation across the contrasts is itself informative, but it is not by itself a statistical test of differences between the three effect estimates. The adjusted individual-galaxy morphology excess is strong against the luminosity-selected cores of richer groups and against true four-member groups, and smallest against the BGG-centred *projected* cores of ordinary groups ($Control_{4C}$), the control definition that most closely mimics a compact projected configuration. This ordering is consistent with a compactness or projected-density sequence — the excess is largest against the two ~ 440 kpc environments ($Control_{4B}$, RG_4) and smallest against the ~ 293 kpc $Control_{4C}$ cores (Sect. 2.1) — and the group-level satellite-composition contrast below shows the same pattern, but it does not prove a causal compactness sequence. It motivates the local projected-density diagnostic of Sect. 3.6 (Fig. 1).

As a compact secondary summary, a pooled model over the deduplicated control catalogues (one row per physical galaxy, cluster-robust by physical Lim group; full family in

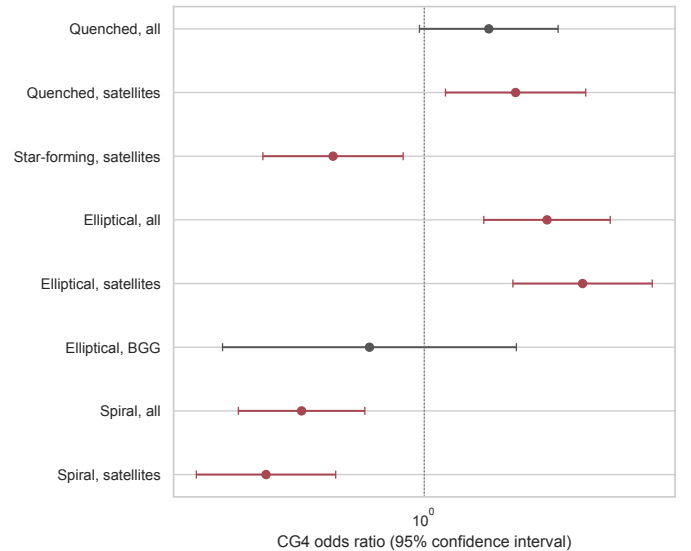


Fig. 2. Adjusted CG_4 odds ratios for star-formation and morphology diagnostics from the pooled secondary models, including all-galaxy, satellite-only, and BGG-only morphology rows. Bars show 95% confidence intervals; the vertical line marks equal odds.

Appendix F gives the same qualitative picture: the pooled elliptical odds ratio is 1.77 (Holm-corrected $p = 8.5 \times 10^{-4}$), and the pooled quenched odds ratio is 1.35 (Holm-corrected $p = 0.136$). The adjusted morphology signal is primarily carried by satellites rather than BGGs: the satellite-only models give an elliptical odds ratio of 2.08 and a spiral odds ratio of 0.48, whereas the BGG elliptical coefficient is not detected after adjustment (OR 0.78; Holm-corrected $p = 0.464$) (Fig. 2).

As announced in Sect. 2.4, all per-control and pooled families were refit without the noisy σ_v covariate (identical samples, clustering, and Holm families; a labelled sensitivity); no model changes Holm-level significance in either direction, and the elliptical odds ratios move by less than their standard errors.

A host-BGG alignment audit supports the satellite interpretation: 56 of 62 non-split CG_4 systems have the same quartet and Lim-host BGG objid, the 6 differences are Pre-dominant systems, the Isolated equalities are trivial quartet=host cases, and all 186 quartet-rank CG_4 satellites are Lim-host satellites. Restricting CG_4 to the 56 BGG-aligned systems leaves all six labelled satellite elliptical/spiral sensitivity contrasts stable; against $Control_{4C}$ the aligned odds ratios are 1.98 and 0.51.

3.3. Matched comparisons

Group-level matched satellite-composition contrast (primary).

Because the four galaxies of a compact group are not independent, the group-level matched estimand treats each quartet as the statistical unit: for each group we count its elliptical-vote satellites ($N_{\text{ell,sat}} \in \{0, 1, 2, 3\}$) and match CG_4 groups one-to-one to control groups on redshift, BGG stellar mass, total quartet luminosity, and velocity dispersion (propensity caliper 0.2SD of the logit). We run the matching three times with the pool restricted to a single control type (one quartet per physical Lim group within each type), and once with the pool drawn from all three types

combined. For the per-control matches, the paired sign-flip permutation p-value is the primary p-value; the bootstrap sign-crossing probability and bootstrap interval quantify robustness and effect-size uncertainty.

The pooled control match covers 54 CG₄ groups and 633 control groups, yielding 54 matched pairs, drawn predominantly from Control_{4B} (46 Control_{4B}, 6 RG₄, 2 Control_{4C}, reflecting the pool composition after deduplication). This pooled match is a secondary summary because the control catalogues overlap. Its elliptical-vote satellite fraction difference is $\Delta = 0.167$ (95% bootstrap interval [0.040, 0.290]; permutation $p = 0.016$, bootstrap $p = 0.010$; $B = 9999$).

The per-control matches (Table 3) give a consistent picture: the CG₄-minus-control fraction difference is positive against all three control types, individually significant without adjustment across control definitions (permutation $p = 0.016$ –0.080) and marginal to significant under the Holm sensitivity of Table F.1. Control_{4C} is the physically most stringent comparison, since its projected-core definition most closely mirrors the CG₄ compactness criterion; against Control_{4C}, the match yields 54 pairs with $\Delta = 0.114$ (95% CI [−0.015, 0.238]; primary permutation $p = 0.080$, bootstrap $p = 0.082$; 0.080 and 0.082 under Holm across the three control definitions, Table F.1). A post-hoc audit of these matches, which condition on group covariates but never on satellite properties, shows the matched pairs are close to balanced in mean satellite stellar mass: the paired CG₄-minus-control difference in mean satellite $\log M_\star$ is 0.03 dex against Control_{4C} (95% CI [−0.08, 0.13]; standardized mean difference 0.09), 0.00 dex against Control_{4B}, and −0.01 dex against RG₄. Repeating each match with mean satellite stellar mass added to the matching covariates estimates a *different quantity* — satellite composition at matched satellite mass rather than at matched group scale — and the two are complementary estimands, not ordered levels of bias correction, because present-day satellite mass may itself encode the accumulated environmental history whose imprint is under test (Sect. 4.1). Under the mass-balanced estimand the Control_{4B} and RG₄ contrasts strengthen ($\Delta = 0.198$ and 0.256; permutation $p = 0.003$ and $p = 3 \times 10^{-4}$) while the already marginal Control_{4C} contrast weakens ($\Delta = 0.086$, permutation $p = 0.202$); both estimands agree that the composition contrast is clear against Control_{4B} and RG₄ and weakest against Control_{4C}. The near-balance in satellite mass is a construction property rather than an accident of the matching: within the shared $\Delta m \leq 3$ window, the distance-ranked Control_{4C} satellites sample the eligible-satellite magnitude-gap distribution of their parent groups nearly democratically (KS $D = 0.04$), as do the CG₄ satellites themselves ($D = 0.06$, $p = 0.44$), whereas the brightness-ranked Control_{4B} selection is strongly weighted toward the bright end ($D = 0.30$, mean satellite $\log M_\star$ higher by 0.21 dex than in CG₄). The galaxy-level match below estimates a different individual-galaxy contrast rather than a weaker version of this group-composition result.

Because σ_v is a noisy matching covariate (Sect. 2.4), all matches were repeated without it: the galaxy-level match retains the same 234 CG₄ galaxies with an unchanged conclusion pattern, and the per-control group-level differences become $\Delta = 0.284$, 0.102, and 0.253 against Control_{4B}, Control_{4C}, and RG₄ (permutation $p = 0.0001$, 0.056, 0.0016) — the same qualitative picture, with Control_{4C} again the weakest contrast.

Table 3. Per-control group-level matched satellite-composition contrasts. Δ is the matched mean difference in elliptical-vote satellite fraction (CG₄ minus control). The paired sign-flip permutation p-value is primary; the bootstrap p-value is the add-one sign-crossing probability used as a robustness check, with $B = 9999$ resamples. P-values are not adjusted across the three control definitions; Appendix F reports the Holm sensitivity. The pooled row is secondary. Pair counts: 54 CG₄ groups have complete matching covariates and classified satellites; against Control_{4B} and Control_{4C} (630 and 632 available control groups) every CG₄ group finds a within-caliper partner, whereas against RG₄ only 52 control groups exist for the 1:1 match without replacement, and the propensity caliper retains 28 pairs on common support.

Control	Pairs	Δ	95% CI	p_{perm}	p_{boot}
Control _{4B}	54	0.160	[0.034, 0.284]	0.016	0.012
Control _{4C}	54	0.114	[−0.015, 0.238]	0.080	0.082
RG ₄	28	0.196	[0.036, 0.351]	0.026	0.019
Pooled	54	0.167	[0.040, 0.290]	0.016	0.010

Galaxy-level matched contrast (secondary). We matched 234 of the 248 available CG₄ galaxies one-to-one to 234 ordinary-group galaxies without replacement, requiring common propensity-score support, a 0.2-standard-deviation caliper and exact rank strata; the control pool is first deduplicated by SDSS objid, no control may share an objid with a CG₄ galaxy, and no control galaxy is used more than once. The propensity model uses $\log M_\star$, redshift, rank, $\log L_{\text{group}}$, and σ_v ; projected distance is retained as a phase-space and interaction diagnostic rather than matched away, because it partly defines compactness and has limited overlap. The largest absolute standardized mean difference changed from 0.37 before matching to 0.08 afterwards; the residual imbalance is carried by $\log L_{\text{group}}$ and lies slightly above the conventional 0.1 threshold, which is why we read the matched contrasts together with the adjusted models rather than in isolation. Matched differences are quoted with group-blocked bootstrap intervals: pairs are resampled by the treated galaxy’s compact group (never galaxy by galaxy), with $B = 9999$ resamples and add-one empirical p-values. Pool composition, provenance, and balance details are given in Appendix F.

The matched CG₄-minus-control difference in elliptical fraction is 0.139, with 95% bootstrap interval [0.037, 0.240] and Holm-corrected $p = 0.036$. Restricted to the satellite pairs — a post-hoc decomposition of this all-pair outcome, reported without family adjustment — the matched elliptical-fraction difference is 0.192 (95% bootstrap interval [0.067, 0.313]; unadjusted $p = 0.004$; 130 complete pairs). The matched quenched-fraction difference is 0.031 ([−0.056, 0.122]; Holm-corrected $p = 1.000$), so the matched data do not establish an enhanced quenched fraction after correction; the star-forming and spiral fractions are exact complements of these outcomes on their complete-case subsets and are reported in Appendix F. Among matched star-forming galaxies, the residual-sSFR difference is −0.147 ([−0.311, 0.008]; $p = 0.064$).

The matched-outcome Holm correction is intentionally conservative: quenched/star-forming and elliptical/spiral diagnostics are retained in the same family even though they are exact complements on their complete-case subsets. Using a non-redundant family would not weaken the morphology conclusion.

The matched individual-galaxy comparison is therefore compatible with the adjusted models but less decisive for morphology than the group-composition estimand; the quenched and star-forming-fraction contrasts are more model-dependent.

is the expected pattern — CG_4 members are preferentially the early types of the dense projected core, so the radius term absorbs the excess — and the positive radius-free coefficient supports the location reading rather than contradicting the conditioned null.

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3.4. Compact-group members versus other members of the same host group

For the Embedded and Predominant CG_4 systems, whose quartets live inside richer Lim parent groups, the sharpest available test of “compact subconfiguration versus shared host environment” compares the CG_4 member galaxies with the *non- CG_4 members of the same host group*. The primary estimator is conditional logistic regression stratified by host group: the stratum-level intercept is conditioned out in the partial likelihood, so every host-level confounder shared by all members of the same host (halo mass, richness, redshift, large-scale environment) is removed by design. As a robustness check we also fit a binomial GLM with host-group fixed-effect dummies and cluster-robust standard errors by host (FE-GLM). The design starts from 54 host groups (958 member galaxies, of which 224 belong to a CG_4); the models include stellar mass, host luminosity rank, and projected host-centric radius as galaxy-level covariates.

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For the elliptical-vote outcome, the complete-case conditional-logit sample contains 791 galaxies in 53 host strata; the remaining design host has no usable Galaxy Zoo elliptical/spiral classification after the model complete-case filters. All 53 complete-case host strata are informative; no concordant strata are dropped. The elliptical-vote odds ratio for CG_4 membership is 1.47 (95% CI [0.92, 2.35]; Holm-corrected $p = 0.216$), and the quenched odds ratio is 0.75 (Holm-corrected $p = 0.269$; 915 complete cases in 54 host strata).

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The FE-GLM yields a similar elliptical-vote point estimate but a smaller estimated uncertainty and nominal significance: OR 1.57 (95% CI [1.02, 2.43]; nominal $p = 0.041$, Holm $p = 0.083$ under the same two-outcome family convention as the conditional logit, so the two estimators agree once the correction is applied like for like). We retain the conditional-logit estimator as primary because it conditions out the host-specific intercept directly.

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At fixed stellar mass, rank, and host-centric radius, no residual excess is detected for compact-subconfiguration membership *within* a shared host under the primary conditional estimator. The confidence interval nevertheless permits a range of moderate effects, so this result does not establish equality. It is consistent with the per-control asymmetry of Sect. 3.2: the CG_4 excess appears more clearly against galaxies of *other* groups than against co-members of the same host, pointing to dense projected-core location rather than compact-group membership alone.

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The radius covariate defines the estimand: conditioning on projected host-centric radius asks whether CG_4 members differ from other members of the same host *beyond their central location*. Refitting the identical complete-case sample and strata *without* the radius term — asking whether they differ *including* location — yields a strong elliptical excess: OR 1.90 (95% CI [1.23, 2.95]; Holm $p = 0.008$) for the conditional logit and 2.06 (Holm $p = 0.0013$) for the FE-GLM, while the quenched outcome remains null in all four specifications (radius-free conditional-logit OR 1.22, $p = 0.41$). Under the local-density interpretation this

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3.5. Morphology robustness

Because Galaxy Zoo morphology is image-based and the adopted vote-fraction cut is one of several defensible choices, we subject the morphology contrast to a battery of robustness tests, summarized in Table 4.

After excluding galaxies whose nearest projected neighbour within the catalogue group lies within 55 arcsec (retaining 200 CG_4 and 2534 $\text{Control}_{4\text{C}}$ galaxies), the CG_4 elliptical fraction *rises* relative to the raw full-sample value in Table 2, rather than falling, and the CG_4 – $\text{Control}_{4\text{C}}$ contrast remains significant in this unadjusted diagnostic. This argues against the simplest interpretation that the elliptical excess is produced solely by crowding in the closest pairs. The close-neighbour flag is itself associated with elliptical-vote morphology (OR 1.98; $p = 6.1 \times 10^{-4}$), so this test does not prove that Galaxy Zoo classifications are immune to crowding; it rules out the closest-pair artefact as the sole origin of the result.

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Because the headline results of Sect. 3.2 are adjusted rather than raw, we additionally refit the full per-control adjusted families on the sample after the same 55-arcsec exclusion, which removes 19% of CG_4 members against 6–10% of the controls (identical covariates and clustering; a separately labelled sensitivity family with its own Holm correction). Every contrast that is Holm-significant in the published families remains so on the adjusted estimand: the all-member elliptical odds ratio moves from 2.14 to 2.16 against $\text{Control}_{4\text{B}}$, from 1.67 to 1.71 against $\text{Control}_{4\text{C}}$, and from 2.81 to 2.71 against RG_4 (Holm $p \leq 0.0044$ in all three cases), with the satellite-only elliptical, spiral-deficit, and $\text{Control}_{4\text{B}}$ quenched families equally stable, and no null result becoming significant. The adjusted excess is therefore not carried by the crowded closest pairs.

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Across classification thresholds of 0.4–0.8 on the deduplicated pooled sample, the adjusted CG_4 elliptical odds ratio varies only mildly and never changes sign (Table 4; the unclassified fraction grows from 13% to 46% over the sweep). A threshold-free check models the debiased elliptical vote fraction p_E itself, and a structural cross-check independent of Galaxy Zoo uses the Simard et al. (2011) pure-Sérsic index (early type: $n_g > 2.5$; 75% coverage); both confirm the direction of the contrast, the latter subject to the differential Simard completeness discussed in Sect. 3.8.

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3.6. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6076 galaxies and refits adjusted models with and without the pairwise tidal index $T_i = \sum_j M_{\star,j}/R_{ij}^3$ (in $\text{M}_\odot \text{kpc}^{-3}$). The sum runs over the galaxy’s *selected quartet co-members only* — not the full Lim host membership, a restriction shown to be innocuous in Appendix F.4 — with no velocity window; R_{ij} converts the great-circle separation with the angular-diameter distance at the quartet’s median redshift (proper kpc); co-members with unmeasured stellar mass contribute zero,

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Table 4. Robustness of the CG₄ morphology contrast. Odds ratios (OR) are adjusted CG₄ coefficients; the continuous row is the adjusted offset in the debiased elliptical vote fraction p_E . P-values in this compact diagnostic table are unadjusted unless explicitly labelled.

Test	Result
Exclude neighbours within 55 arcsec	elliptical fraction 52.5% (CG ₄) vs 40.1% (Control _{4C}); $p = 6.2 \times 10^{-4}$
Close-neighbour-flag model	OR 1.76 [1.30, 2.39]; $p = 2.6 \times 10^{-4}$
Threshold sweep 0.4–0.8	OR 1.73–2.19
Continuous p_E offset	0.089 [0.047, 0.131]; $p < 10^{-4}$
Sérsic $n_g > 2.5$ cross-check	OR 1.60; $p = 0.002$

making T_i a lower bound in those groups, and quartets with any member lacking coordinates or redshift are dropped whole. These exploratory models adjust for stellar mass and satellite/BGG status only, pool the three control catalogues *without* objid deduplication (unlike the pooled models of Appendix F), and cluster standard errors by physical group. The two model variants estimate *different quantities*. The model without the index estimates the *total* association between compact-group membership and the outcome at fixed stellar mass and satellite status; the model with the index estimates the association *at fixed projected local density*. Because the index is built from R^{-3} -weighted projected separations, it is mathematically entangled with the compact-group selection itself (compact groups are selected to have small projected separations), so conditioning on it partially conditions on the exposure. A reduction of the CG₄ coefficient when the index is added is thus *conditional attenuation*, not mediation: it does not show that local density accounts for the morphology contrast, and it cannot, because the causal ordering of selection, density, and morphology is not identified by projected snapshots. The baseline models in this subsection are refit on the pairwise-analysis complete-case sample, so their coefficients need not be numerically identical to the corresponding coefficients in Sect. 3.2 or Appendix F; because R_{ij} is projected, the term is a local projected-density and immediate-neighbour proxy rather than a direct tidal-history measurement.

For elliptical morphology, the baseline CG₄ odds ratio is 1.95 ($p < 10^{-4}$), whereas the model including the tidal index gives a CG₄ odds ratio of 1.39 ($p = 0.031$); the tidal-index term itself has odds ratio 1.42 per 1.07-dex standard deviation of $\log_{10} T_i$ ($p < 10^{-4}$). The projected tidal-density proxy therefore *attenuates* the CG₄ coefficient; given the selection entanglement described above, this conditional attenuation must not be read as the index accounting for the effect, and the result remains an exploratory projected-pair diagnostic.

For quenched status, the baseline CG₄ odds ratio is 1.48 ($p = 0.014$), while the model including the tidal index gives 1.02 ($p = 0.928$), with a tidal-index term of odds ratio 1.51 ($p < 10^{-4}$).

Common-support diagnostics quantify how far the index can be conditioned on. The CG₄ and control $\log T_i$ distributions have an overlap coefficient of 0.54, but their medians differ by 1.1 dex (per control: +1.34 dex versus Control_{4B}, +0.81 versus Control_{4C}, and +1.22 versus RG₄) and 15% of CG₄ galaxies lie above the pooled-control 95th percentile of $\log T_i$; collinearity in the adjusted design is nevertheless mild (point-biserial $r = 0.18$; variance-inflation

factor of the CG₄ indicator 1.03). Sensitivity refits reflect that thin far tail: the elliptical CG₄ odds ratio at fixed T_i is 1.39 in the linear fit above, 1.40 with a natural cubic spline in $\log T_i$, 1.27 after deduplicating the control pool by objid ($p = 0.12$), and 1.22 ($p = 0.20$) when trimmed to the common-support window of $\log T_i$. The residual CG₄ coefficient at fixed projected density is therefore not stable enough to bear physical weight, reinforcing the conditional-attenuation reading above.

3.7. Secondary outcomes

Star formation. The adjusted and matched analyses above already bound the star-formation contrast: the per-control quenched odds ratios are control-dependent (Sect. 3.2), the matched quenched and star-forming fractions show no detected offset after correction (Sect. 3.3), and the main-sequence residuals are a secondary diagnostic (Sect. 3.1). The largest per-control main-sequence offset is against Control_{4C} ($\Delta = 0.058$, 68% interval 0.039–0.103; $p = 0.024$; Appendix D), whereas the matched residual-sSFR contrast, which pools the control catalogues, shows no offset (−0.147; $p = 0.064$; Sect. 3.3); the per-control comparison is unadjusted and is not read as a detection.

The interpretation does not rely on the exact GMM division of the sSFR plane: repeating the key satellite comparison with a fixed threshold, $\log_{10}(\text{sSFR}/\text{yr}^{-1}) = -11.0$, gives CG₄ and RG₄ satellite star-forming fractions of 41.4% and 60.2%, respectively, a difference of −18.8 percentage points ($p = 5.3 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the quenched/star-forming contrast remains more sensitive to modelling choices.

H α emission. Our retrieved catalogues do not include the age-sensitive indices D_n4000 and $\text{H}\delta_A$ (available in the MPA-JHU `galSpecIndx` product), so we defer a post-starburst classification to follow-up work and perform only a limited H α -equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission and classifying strong H α emission by $\text{H}\alpha \text{EW} \leq -3 \text{ \AA}$.

The strong-emission fraction is 32.3% in CG₄, 41.0% in Control_{4B}, 43.3% in Control_{4C}, 56.9% in RG₄.

Relative to Control_{4B}, the CG₄ strong-emission fraction differs by −0.087 (Holm-corrected $p = 0.012$).

Relative to Control_{4C}, the CG₄ strong-emission fraction differs by −0.110 (Holm-corrected $p = 0.003$).

Relative to RG₄, the CG₄ strong-emission fraction differs by −0.246 (Holm-corrected $p < 10^{-4}$).

These strong-H α fractions are reported as *descriptive* diagnostics only: they are not adjusted for the samples' different morphology mixes, and since LINER-like emission is common in early-type galaxies, a morphology-blind emission fraction partially restates the morphology contrast itself. They indicate a lower strong-emission fraction in CG₄ than in each control, but provide neither a direct post-starburst or recently-quenched classification nor an independent star-formation test.

Emission-line and AGN-like populations. The emission-line analysis classifies BPT star-forming, composite, and

AGN-like populations for galaxies with the required diagnostic line ratios: composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations in the [NII] BPT plane, and AGN-like galaxies lie above the Kewley et al. (2001) line; galaxies with a pre-existing AGN flag but no BPT line ratios are also counted as AGN-like, and BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG₄ galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG₄ AGN-like odds ratio is 1.17 ($p = 0.364$): AGN-like activity does not provide a robust independent compact-group discriminator.

Magnitude gaps and fossil-like assembly. We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap; large gaps are the classic optical signature of fossil-like assembly (Ponman et al. 1994; Jones et al. 2003; Dariush et al. 2010).

For Δm_{12} , the CG₄ and control medians are 1.17 and 0.99 mag, respectively (Holm-corrected $p = 0.027$); for Δm_{14} , they are 2.56 and 2.37 mag (Holm-corrected $p = 0.007$).

The correlation between Δm_{12} and quenched satellite fraction is $\rho_S = -0.18$ (Holm-corrected $p < 10^{-4}$).

Projected phase space. We ask whether the CG₄–RG₄ satellite contrasts vary with projected position relative to the BGG. The analysis uses 174 CG₄ and 166 RG₄ satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and, when available, uses $|\Delta v_{\text{BGG}}|/\sigma_v$ as a projected phase-space diagnostic; such coordinates are statistical proxies for infall state or quenching time, not orbital reconstructions (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG₄ and 48.2% for RG₄, a difference of 20.1 percentage points; the corresponding early-type fractions are 64.7% and 35.6% (difference 29.1 points).

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG₄ offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG₄ odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$), and the analogous early-type model gives 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check on 126 CG₄–RG₄ satellite pairs gives matched quenched- and early-type-fraction differences of 9.5 and 18.7 percentage points, respectively.

In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites, so we treat these bins as an observational diagnostic of where the contrast appears, not as a dynamical clock.

Optical colours. We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry and, because stellar mass confounds colour comparisons (Alpaslan et al. 2015), model each colour jointly with $\log(M_\star/M_\odot)$ and redshift, with cluster-robust standard errors. The colour-

analysis subset is not representative of the full galaxy catalogues – matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies – so colours are treated as a secondary diagnostic. In the pooled satellite comparison, no blue CG₄ coefficient is detected after Holm correction, and the adjusted coefficients are control-sample dependent. The data therefore do not establish a robust global blue or red offset for CG₄ satellites; the full colour–mass, model-dependence, and radial diagnostics are given in Appendix E.6.

Group-scale diagnostics. A broader set of environmental diagnostics is reported in Appendix E: significant monotonic sSFR–distance trends in individual samples (Appendix E.1); dominance splits whose clearest separation is in group luminosity rather than in any CG₄-specific property; morphology tests against the Zheng–Shen class, Δm_{12} , and the BGG luminosity fraction, none of which survives false-discovery-rate control; and a crossing-time diagnostic that primarily tracks virial mass and mass-to-light ratio.

3.8. Galaxy sizes at fixed stellar mass

We now test whether CG₄ galaxies differ in size at fixed stellar mass, using the seeing-corrected Simard half-light radii as the primary measure and the Petrosian radii as the model-independent cross-check (Sect. 2.3). The Simard measure resolves 77.4% of the CG₄ galaxies and 71.1% of the pooled control galaxies, while the Petrosian radius is essentially complete in every sample (Appendix Fig. F.2). Because the CG₄–control difference in Simard completeness exceeds five percentage points, every Simard-based coefficient below carries a differential-completeness caveat and is read together with the Petrosian rerun, which does not share this incompleteness.

For satellite galaxy size, the CG₄ offset at fixed stellar mass, redshift, and available group-scale covariates is -0.010 dex (-2.3%), with 95% confidence interval $[-0.039, 0.019]$ and Holm-corrected $p = 1.000$, over 2575 satellites in 754 physical groups.

The all-galaxy model gives -0.006 dex (95% CI $[-0.032, 0.020]$; Holm-corrected $p = 1.000$), and the BGG-only model gives 0.019 dex (Holm-corrected $p = 1.000$).

No coefficient in this primary family is detected after Holm correction, so the pooled adjusted models do not establish a CG₄ size offset at fixed stellar mass.

The per-control versions of the fit are descriptive (Appendix Fig. G.1); the fitted CG₄ coefficient is -0.035 dex against Control_{4B}, 0.000 dex against Control_{4C}, -0.006 dex against RG₄ (raw p-values corrected with Benjamini–Hochberg within a family defined as exactly the per-control size coefficients displayed in Fig. G.1 — the three control contrasts of each size measure shown — and containing no test outside that figure). The sign and significance of the CG₄ coefficient depend on the adopted control, so we do not interpret any single per-control contrast on its own.

The one-to-one matched comparison re-derives the same 234 propensity pairs as the matched-control analysis, with a separate Holm family for the three size outcomes so that the published matched-outcome family is untouched. On the 166 pairs in which both members have valid Simard sizes, the mean CG₄–minus-control difference in $\log_{10} R_{\text{chl},r}$ is

−0.019 dex (95% bootstrap interval [−0.064, 0.026]; Holm-corrected $p = 0.800$). The corresponding Petrosian difference is −0.033 dex (Holm-corrected $p = 0.089$), and the matched concentration difference is −0.008 (Holm-corrected $p = 0.904$).

The Petrosian rerun, with the field seeing as an additional standardized covariate, gives a satellite coefficient of −0.019 dex (Holm-corrected $p = 0.332$) and an all-galaxy coefficient of −0.012 dex (Holm-corrected $p = 0.332$).

The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.068 (Holm-corrected $p = 0.332$), so no CG₄ concentration-index shift is detected at fixed mass.

Crowding, measure-difference, tidal-index, radial, and morphology-stratified size diagnostics are given in Appendix G.

4. Discussion: drivers of structural evolution in compact groups

4.1. The morphology hierarchy and its estimands

The clearest residual difference between CG₄ and the regular-group controls is morphological: a higher Galaxy Zoo 1 elliptical-vote fraction and a lower spiral-vote fraction, carried primarily by satellites. Its strength is graded by control construction and estimand. At the galaxy level, the per-control adjusted excess is strong against Control_{4B} and RG₄ and smallest against the projected cores Control_{4C}; at the group level, the matched satellite fraction is higher in CG₄ than in every control (Table 3), individually significant against Control_{4B} and RG₄ and marginal against Control_{4C}. This ordering is consistent with compactness or projected density mattering, but it does not establish a causal compactness sequence or a statistically demonstrated difference between the three control-specific estimates. The galaxy-level matched contrast is directionally consistent but not detected once groups are resampled together, and the primary within-host estimator detects no additional residual excess over co-members of the same host.

These results are not contradictory: the analyses differ in matched populations, conditioning sets, outcomes, effect scales, and uncertainty units. The satellites-only decomposition of Sect. 3.3 also shows BGG dilution: restricted to satellite pairs, the matched difference rises to 0.192, close to the group-level per-control differences. The group-level analysis estimates system-level satellite composition, whereas the galaxy-level analyses estimate conditional individual-galaxy contrasts; the odds-ratio scale is also non-collapsible. We therefore read the pattern as estimand dependence: the morphology contrast is visible at both levels, smallest for the control closest to dense projected cores, and absent within shared hosts once central location is conditioned away. A fixed-mass caution remains: present-day stellar mass may encode part of the environmental history under test, so contrasts at matched or adjusted M_* estimate the residual beyond that history, not the total historical effect.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical morphology, the pairwise complete-case baseline CG₄ odds ratio is 1.95 ($p < 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG₄ odds ratio decreases to 1.39

($p = 0.031$), while the tidal-index term itself has odds ratio 1.42 ($p < 10^{-4}$). This behaviour is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the elliptical/early-type-like population.

This proxy uses projected separations, so it may be inflated by line-of-sight alignments and is entangled with compact-group selection itself. We therefore read the tidal-index result as a local projected-density or immediate-neighbour proxy, not as a time-dependent tidal-field measurement. The Control_{4C} comparison is stringent because Control_{4C} is also selected as a projected compact core around a BGG. The persistence of a CG₄–Control_{4C} morphology contrast after closest-neighbour exclusion, together with attenuation in the tidal-index model, is consistent with CG₄ selecting a more extreme part of the local-density distribution rather than establishing a separate causal channel.

This reading is consistent with compact groups tracing the high-density, high-interaction tail of group environments – some genuinely compact, others dense cores or substructures inside larger haloes – and with stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work. The 6 isolated CG₄s are compatible with loose RG₄ groups in satellite early-type fraction (group-level permutation $p = 1.00$) and have a lower satellite early-type fraction than the rest of CG₄ ($p = 0.023$); this pattern is consistent with a larger-scale host contribution, but the very small isolated sample prevents a firm separation of explanations (Appendix E). The available M_{200c} is a host-halo abundance-matching mass rather than a harmonised quartet property; adding it to the adjusted models mildly attenuates the elliptical odds ratios and changes no conclusion.

4.2. Elliptical-vote morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo variables identify high elliptical and combined-spiral vote fractions; they do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other bulge-dominated systems. The enhanced CG₄ elliptical fraction should therefore be read as an early-type-like appearance shift, not proof that compact groups specifically build classical ellipticals.

This caveat matters when comparing with recent S-PLUS studies of compact-group transition galaxies and R_e –Sérsic-index trends (Montaguth et al. 2023, 2025, 2026a,b). Their transition population can plausibly overlap with our elliptical-vote class, so the approaches are complementary: their structural measurements describe likely transformation channels, while our regular-group controls identify the residual elliptical/spiral vote-fraction balance. The attached Simard R_e –Sérsic- n plane is therefore shown as context, not as a primary test (Appendix Fig. G.2).

4.3. Why the star-formation and size signals are weaker

The observed star-formation contrast is consistent with being carried by the differing vote-based morphology mix rather than by an additional conditional offset. At fixed vote-based morphology, the conditional quenched fractions do not differ detectably across samples ($P(Q|E) = 0.908$,

0.903, 0.887, and 0.850 for CG_4 , $\text{Control}_{4\text{B}}$, $\text{Control}_{4\text{C}}$, and RG_4 ; homogeneity $p = 0.39$). A Kitagawa decomposition leaves conditional terms consistent with zero: -0.034 $[-0.086, +0.019]$ versus $\text{Control}_{4\text{B}}$, -0.016 $[-0.068, +0.039]$ versus $\text{Control}_{4\text{C}}$, and $+0.035$ $[-0.047, +0.121]$ versus RG_4 (group-blocked 95% intervals; Table B.2). Thus the quenched odds-ratio pattern is statistically attributable to the morphology mix in this observational decomposition, with no additional quenching offset detected at fixed vote-based morphology.

The star-formation non-detections are also limited by the spectroscopy. CG_4 has more catalogued galaxies with a nearest neighbour below the 55-arcsec fibre-collision scale than the controls (19.4% versus 7.7%). $\text{Control}_{4\text{C}}$ is higher still (9.9%) because it is selected as a compact-core analogue. Additional close companions may therefore be preferentially missing in CG_4 , underrepresenting short-lived starbursts or post-starburst phases and biasing the pairwise tidal index downward. The MPA-JHU aperture corrections may also be less secure for interacting or blended members (Brinchmann et al. 2004; Kauffmann et al. 2003a). We therefore treat the sSFR, SFMS, and $\text{H}\alpha$ results as conservative secondary non-detections, not as proof of no star-formation effect.

The size test carries a parallel message. Coenda et al. (2012) found compact-group galaxies to be smaller at fixed luminosity than loose-group and field samples, especially for early types, using SDSS Petrosian half-light radii. We re-pose that question with the non-split HMCG-based CG_4 sample against three regular-group controls, seeing-corrected model sizes, a Petrosian cross-check, cluster-robust adjusted models, one-to-one matching, crowding tests, and Holm-corrected intervals.

Under this stricter design, we find no compact-group size offset at fixed stellar mass relative to regular groups; the primary satellite fit bounds any mean CG_4 size deficit to roughly 0.04 dex at the 95% level. This does not contradict Coenda et al. (2012) directly, because their comparison was against loose groups and the field, whereas ours is against regular-group cores selected from the same parent catalogues.

Two guardrails apply: a negative Δ at fixed mass can reflect tidal truncation or stripping, but also higher bulge dominance or residual blending in crowded fields. The morphology-stratified fits address the former, and the Petrosian cross-check, measure-difference diagnostic, and crowding exclusions bound the latter (Appendix G); future claims of compact-group size evolution should be read through these blocks together, not through a single pooled coefficient.

5. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems, comparing them with RG_4 , $\text{Control}_{4\text{B}}$, and $\text{Control}_{4\text{C}}$ through the individual-galaxy, group-composition, and shared-host contrasts defined in Sect. 2.4. The inference applies to this luminous, volume-limited, exactly four-member non-split HMCG sample rather than to all compact groups.

The main result is morphological and graded by control construction. CG_4 galaxies – primarily satellites – have higher Galaxy Zoo 1 elliptical-vote and lower spiral-vote fractions. The adjusted individual-galaxy excess is

strong against $\text{Control}_{4\text{B}}$ and RG_4 and smallest against $\text{Control}_{4\text{C}}$; matched group-level satellite-composition contrasts are positive against all three controls, individually significant against $\text{Control}_{4\text{B}}$ and RG_4 and marginal against $\text{Control}_{4\text{C}}$, with the same hierarchy in satellite-mass-balanced variants. This ordering is consistent with compactness or projected density mattering, but it does not prove a causal compactness sequence. No additional residual excess is detected within shared hosts. The morphology result is stable to vote-threshold, vote-fraction, Sérsic-index, and close-pair checks; BGG morphology is not detected as different after adjustment. Conditioning on a local projected-density proxy attenuates the galaxy-level coefficient, but this conditional attenuation does not establish mediation.

The star-formation result is weaker and statistically attributable to the differing morphology mix, with no additional conditional offset detected at fixed vote-based morphology. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4 , but adjusted evidence is control-dependent, matched contrasts provide no evidence for a quenched-fraction difference, and star-forming CG_4 galaxies show no robust main-sequence offset.

At fixed stellar mass, redshift, and group-scale covariates, no CG_4 offset is detected in seeing-corrected half-light radius, in either the adjusted models or the matched pairs; the primary satellite fit bounds any mean size deficit to roughly 0.04 dex.

The remaining secondary diagnostics do not identify a simple alternative driver: colours are control-sensitive and based on a biased matched subset, AGN-like fractions and magnitude gaps are not detected as parallel offsets, the $\text{H}\alpha$ diagnostic lacks age-sensitive spectral indices, and projected phase-space trends do not localize the contrast uniquely to the innermost satellites.

We therefore interpret CG_4 systems as hosting a population shifted towards early-type-like morphologies, consistent with accumulated structural transformation in dense group cores and/or preprocessing. This does not establish a compact-group-specific quenching mechanism at the present epoch, and the dynamically isolated subset is too small to isolate a distinct mode. Explicit E–S0–spiral and transition-galaxy classifications, bulge–disc structure, homogeneous halo masses, and age-sensitive stellar-population diagnostics will be needed to determine whether the satellite excess reflects disk fading, tidal heating, or merger-driven transformation.

Data availability

The input data are public SDSS, MPA-JHU, Galaxy Zoo, Simard et al. (2011), group, and compact-group catalogues cited above. Derived galaxy and group tables will be made available through CDS at publication. The analysis and paper-generation code is available at https://github.com/MatthieuTricottet/CG_Gals; the article release, including online diagnostic supplements, will be archived with a DOI at acceptance.

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2007), statsmodels (Seabold & Perktold 2010), and scikit-learn (Pedregosa et al. 2011).

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Appendix A: SDSS data query

We specifically extract the columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table. In the query, `galSpecExtra`, `galSpecLine`, and `zooSpec` are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`; the queried quantities are then merged onto the `CG4` and control catalogue rows by their stored `objid`.

The full query is provided in `src/data_loader.py` in the code repository (see Data availability).

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha) \quad \text{and} \quad \log_{10}([\text{O III}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10} \frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of [Kauffmann et al. \(2003a\)](#):

$$\log_{10} \frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 3.7, in which composite galaxies lie between the [Kauffmann et al. \(2003a\)](#) and [Kewley et al. \(2001\)](#) demarcations and only galaxies above the [Kewley et al. \(2001\)](#) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the emission-line/AGN-like analysis treats BPT-unclassified objects separately. Galaxies with `specsfr_tot_p50` equal to `-9999` in the SDSS carry the MPA-JHU missing-value sentinel and have no valid sSFR estimate; they are treated as missing data throughout: excluded from the GMM calibration, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_\star$ - $\log \text{sSFR}$ plane ([McLachlan & Peel 2000](#)). The calibration excludes galaxies flagged as AGN-like where the required BPT line ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and sSFR estimates can still receive a broad sSFR class. The quenched/star-forming boundary is the locus where the two component posterior probabilities are equal; the resulting classification for the SDSS selection is shown in Fig. B.1. A fixed-threshold robustness check confirms that the qualitative hierarchy is not tied to the exact GMM split (Sect. 3.7).

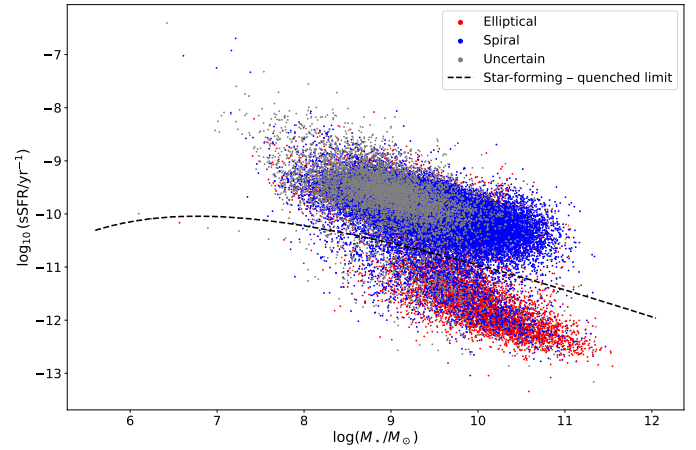


Fig. B.1. sSFR versus stellar mass, the star-forming/quenched limit, and Galaxy Zoo morphologies for the SDSS selection, excluding galaxies without sSFR estimates.

Table B.1. Extreme-assignment bounds on the raw `CG4`-minus-control quenched-fraction difference. “Missing” columns give the missing-sSFR percentage of each side; the observed difference uses classified galaxies only; min/max are the extreme corners with all missing galaxies on each side assigned to quenched or star-forming (all-galaxy denominators).

Scope	Control	Missing (%)	Δf_Q^{obs}	min	max
		<code>CG₄</code>	ctrl		
All	Control _{4B}	7.3	6.4	0.025	-0.046 0.091
All	Control _{4C}	7.3	6.8	0.040	-0.034 0.107
All	RG ₄	7.3	2.2	0.174	0.115 0.210
Sat.	Control _{4B}	6.5	5.8	0.043	-0.021 0.101
Sat.	Control _{4C}	6.5	6.3	0.063	-0.005 0.122
Sat.	RG ₄	6.5	1.2	0.201	0.156 0.233

Extreme-assignment bounds for the missing class. Because the missing-sSFR class could in principle hide differential quenching, Table B.1 bounds every raw quenched-fraction contrast by the four extreme assignments in which *all* missing-sSFR galaxies on *each* side are set to quenched or to star-forming, with consistent all-galaxy denominators on both sides (so `CG4` is never compared with a classified-only control fraction). The differential missingness is negligible against Control_{4B} and Control_{4C} (7% of `CG4` members versus 6% and 7%) and material only against RG₄ (2%). The raw RG₄ contrast survives all four corners (all-member difference between 0.12 and 0.21 around the observed 0.17; satellites 0.16 to 0.23), whereas the small Control_{4B} and Control_{4C} differences can change sign only under the doubly adversarial corner and are not the basis of any claim in this paper.

Star formation at fixed morphology. Table B.2 gives the quenched fractions among sSFR-classified galaxies at fixed vote-based morphology. The four samples are statistically compatible at fixed morphology (homogeneity of $P(Q|E)$: $\chi^2 p = 0.39$); the Kitagawa decomposition quoted in Sect. 4.3 splits each raw `CG4`-minus-control quenched-fraction offset into its morphology-mix and conditional parts on this basis, with group-blocked bootstrap intervals (4000 draws).

Table B.2. Conditional quenched fractions $P(Q|\text{morphology})$ among sSFR-classified galaxies, with Wilson 95% intervals.

	CG ₄	Control _{4B}	Control _{4C}	RG ₄
Elliptical	0.908 ^{+0.040} _{-0.065}	0.903 ^{+0.016} _{-0.019}	0.887 ^{+0.017} _{-0.020}	0.850 ^{+0.069} _{-0.111}
Spiral	0.326 ^{+0.105} _{-0.090}	0.357 ^{+0.027} _{-0.026}	0.328 ^{+0.027} _{-0.026}	0.295 ^{+0.084} _{-0.072}

Notes. For the uncertain-morphology class, $P(Q|U) = 0.375, 0.601, 0.608, \text{ and } 0.400$ ($n = 24, 258, 288, \text{ and } 30$ classified galaxies, same sample order).

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 61 of the 765 parent Lim groups, the would-be Control_{4C} quartet (BGG plus the three closest projected companions within 3 magnitudes) contains at least one CG₄ galaxy — 233 CG₄ galaxies in total. Table C.1 breaks this overlap down by the Zheng & Shen (2021) embeddedness class: essentially every Embedded and Predominant compact group *is* the projected core of an ordinary group. The Isolated entries are trivial self-overlap: for each of the 6 isolated systems, the overlapping Lim group is the compact group's own host, whose BGG and three nearest projected companions are exactly the four CG₄ members. Had these groups not been excluded, the same galaxies would have appeared on both sides of the comparisons, biasing every contrast toward null.

Appendix D: Star-forming main-sequence details

We model the star-forming main sequence as a function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is provided in the online supplement. The diagnostic shown there is the residual RMS scatter, not a formal reduced χ^2 , because no per-galaxy uncertainty model is used. We adopt order 2 because it is the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.305 dex, compared with 0.306 dex for order 1). We then use order 2 to compute the residual of each star-forming galaxy with respect to the main sequence. A positive residual means higher sSFR at fixed stellar mass relative to the fitted star-forming main sequence.

In the comparisons below, Δ is the CG₄-minus-control median residual; positive values mean that CG₄ star-forming galaxies lie above the fitted main sequence relative to the control. Each quoted interval is the central 68% (16th–84th percentile) bootstrap interval on the median difference, and each p is the two-sided bootstrap sign-crossing

Table C.1. CG₄ galaxies inside would-be Control_{4C} quartets of the parent catalogue, by compact-group class.

CG ₄ class	CG ₄ groups	Lim groups	Galaxies
Embedded	37 / 37	37	145
Isolated	6 / 6	6	24
Predominant	14 / 19	14	52
Split	4 / 16	4	12

Notes. “CG₄ groups” counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

probability for the same difference (10 000 resamples). The 68% interval and the two-sided p answer slightly different questions and can therefore appear to disagree near the significance boundary: a 68% interval that excludes zero is compatible with p slightly above 0.05.

The median main-sequence residual of CG₄ galaxies is:

- marginally different from that of Control_{4B}, with a median offset of $\Delta = 0.056$ ($0.028 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.056$.
- significantly different from that of Control_{4C}, with a median offset of $\Delta = 0.058$ ($0.039 \leq \Delta \leq 0.103$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.024$.
- not significantly different from that of RG₄, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.532$.

Because the raw residual comparison and the matched comparison condition on different covariate structures, and because the apparent offset changes in strength after matching, we treat SFMS residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset (Sect. 3.1).

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

Appendix E: Environmental, group-scale, and colour diagnostics

This appendix collects the group-scale and colour diagnostics summarized in Sect. 3.7.

E.1. Distance to the BGG

We compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 174 satellites ($p = 1.5 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 5.5 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1972 satellites ($p = 3.7 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.4 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.19$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.16$ ($p < 10^{-4}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.11$ for 1977 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.11$ ($p < 10^{-4}$).

E.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity,

and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG₄ shows no significant dominated/non-dominated difference after the Benjamini–Hochberg false-discovery-rate (FDR) correction. Here p_{adj} denotes the FDR-adjusted p -value, and Cliff’s δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower L_{group} than their non-dominated counterparts: Control_{4B} gives 9.1×10^{10} versus $1.28 \times 10^{11} L_{\odot}$ ($p_{\text{adj}} < 10^{-4}$, $\delta = -0.55$); Control_{4C} gives 9.66×10^{10} versus $1.07 \times 10^{11} L_{\odot}$ ($p_{\text{adj}} < 10^{-4}$, $\delta = -0.25$); and RG₄ gives 7.09×10^{10} versus $9.82 \times 10^{10} L_{\odot}$ ($p_{\text{adj}} = 1 \times 10^{-4}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control_{4C}, where S_{frac} increases with $\langle R_{ij} \rangle$ in both dominated and non-dominated groups ($\rho_S = 0.29$, $p < 10^{-4}$ and $\rho_S = 0.29$, $p < 10^{-4}$), while weaker negative trends are found with σ_v and L_{group} . Control_{4B} contributes only weaker non-dominated-group correlations. In CG₄, the significant trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_S = -0.50$, 36 groups) and with $\Delta V_{\text{BGG}}/\sigma_v$ among non-dominated groups ($\rho_S = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional diagnostics are provided in the online supplement.

E.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in the following cases.

- Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.842 ($p = 0.01$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.842 ($p = 0.02$).
- Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.972 ($p < 10^{-4}$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.972 ($p < 10^{-4}$).

In each case the group-level and satellite-level clustered treatments share the same point estimate by construction and differ only in the uncertainty model, which is why the odds ratios coincide.

E.4. Morphology versus Zheng–Shen class, magnitude gap, and BGG dominance

We next test whether the E/Sp morphology mix is associated with three group-level descriptors: the Zheng–Shen compact-group class, the first–second magnitude gap $\Delta m_{12} = M_{r,2} - M_{r,1}$, and the continuous BGG luminosity

Table E.1. Elliptical and spiral counts by Zheng–Shen class for the CG₄ sample. Uncertain morphologies are excluded. Split systems have zero counts because they were removed by the existing non-split CG₄ sample definition.

Zheng–Shen class	All CG ₄ E/Sp	CG ₄ BGG E/Sp	CG ₄ sat. E/Sp
Split	0/0	0/0	0/0
Isolated	7/11	3/1	4/10
Predominant	41/22	11/5	30/17
Embedded	76/53	22/11	54/42

Table E.2. Elliptical fractions $f_E = N_E/(N_E + N_{\text{Sp}})$ and median host halo mass by Zheng–Shen class, with Wilson 95% intervals.

Class	Groups	f_E (all)	f_E (sat.)	$\log M_{200c}$
Isolated	6	$0.389^{+0.225}_{-0.186}$	$0.286^{+0.261}_{-0.169}$	12.86
Embedded	37	$0.589^{+0.081}_{-0.086}$	$0.562^{+0.095}_{-0.100}$	13.13
Predominant	19	$0.651^{+0.106}_{-0.123}$	$0.638^{+0.122}_{-0.143}$	13.83

Notes. Host $\log M_{200c}$ is the Yang–Lim abundance-matching host-halo mass (Appendix F.4); for isolated systems it describes the quartet itself.

fraction $f_{L,\text{BGG}} = L_{\text{BGG}}/L_{\text{group}}$. These tests use only the existing Galaxy Zoo elliptical and spiral classes; uncertain morphologies are excluded from the binary fits. We run the tests separately for all galaxies, BGGs only, and satellites only, so that any trend driven by the BGG itself is not folded into the satellite result. For satellites, $f_{L,\text{BGG}}$ mechanically includes the satellite luminosity in the group-luminosity denominator, so any satellite-level trend with $f_{L,\text{BGG}}$ is partly definitional; we therefore report the satellite models with a stellar-mass control.

The magnitude-gap and BGG-luminosity-fraction quantities are available for all 118 CG₄+RG₄ groups used in this test. The Zheng–Shen class is available only for compact groups. Split systems are absent from the current CG₄ sample because they are removed before the paper analysis, and no corresponding Zheng–Shen class is defined for the RG₄ catalogue used here. We therefore report the RG₄ class test as unavailable, while retaining RG₄ in the Δm_{12} and $f_{L,\text{BGG}}$ comparisons.

Overall, these tests provide little evidence that the E/Sp mix is driven by Zheng–Shen class, magnitude gap, or BGG luminosity fraction. The few marginal associations are therefore treated as descriptive diagnostics rather than as robust secondary trends (Tables E.1 and E.3).

For the Zheng–Shen compact-group class, no CG₄ association reaches the conventional 5% significance threshold. The closest case is the satellite-only test, which is treated as marginal and descriptive. Because the Zheng–Shen class is not defined for the RG₄ catalogue, we do not attempt an RG₄ class comparison.

The class structure of the early-type fraction itself is summarized in Table E.2. Group-level permutation tests on the difference of per-group mean satellite f_E (20 000 permutations, two-sided, add-one convention) give $p = 0.023$ for Isolated versus the rest of CG₄, $p = 1.00$ for Isolated versus RG₄, and $p = 0.40$ for Isolated versus Control_{4C}: the isolated systems are compatible with the loose RG₄ groups and sit below the embedded classes of their own sample.

For Δm_{12} , no statistically significant morphology trend is found in CG₄ or RG₄, whether the fit is run for all

Table E.3. Tests of the E-vs-Sp morphology mix against compact-group class, magnitude gap, and BGG luminosity fraction. The continuous columns report the available adjusted logistic model for $P(E)$; all-galaxy and satellite fits use group-clustered standard errors. Odds ratios are reported per magnitude for Δm_{12} and per 0.1 increase in luminosity fraction for $f_{L,BGG}$.

Subset	Sample	Test	Result
all galaxies	CG ₄	Class	$p = 0.137$
		Δm_{12}	OR 1.42, $p = 0.167$
		$f_{L,BGG}$	OR 1.29, $p = 0.080$
all galaxies	RG ₄	Class	not defined
		Δm_{12}	OR 1.33, $p = 0.220$
		$f_{L,BGG}$	OR 1.09, $p = 0.540$
BGGs only	CG ₄	Class	$p = 1.000$
		Δm_{12}	OR 1.77, $p = 0.426$
		$f_{L,BGG}$	OR 1.68, $p = 0.174$
BGGs only	RG ₄	Class	not defined
		Δm_{12}	OR 1.58, $p = 0.312$
		$f_{L,BGG}$	OR 1.21, $p = 0.520$
satellites only	CG ₄	Class	$p = 0.066$
		Δm_{12}	OR 1.28, $p = 0.346$
		$f_{L,BGG}$	OR 1.18, $p = 0.321$
satellites only	RG ₄	Class	not defined
		Δm_{12}	OR 1.41, $p = 0.342$
		$f_{L,BGG}$	OR 1.12, $p = 0.584$

galaxies, BGGs only, or satellites only. The same conclusion holds for $f_{L,BGG}$: no fitted trend reaches $p < 0.05$. The CG₄-RG₄ interaction models detect no significant difference in the morphology–dominance relation for either Δm_{12} or $f_{L,BGG}$, across all galaxies, BGGs only, and satellites only. A class interaction is not defined because the Zheng–Shen compact-group class has no RG₄ counterpart in the present catalogue. Across the morphology–class, morphology–gap, morphology–luminosity-fraction, and interaction tests, no result remains significant after Benjamini–Hochberg FDR control.

E.5. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(M_{\text{VT}}/L_r)$ and $\log(M_{\text{VT}})$ are $\rho_S = -0.39$ ($p < 10^{-4}$) and $\rho_S = -0.33$ ($p < 10^{-4}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally detected but very weak ($\rho_S = 0.08$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group} .

E.6. Optical colours

The retained colour-analysis matches include 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1153 of 2812 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. This stricter subset differs from the availability audit in Fig. F.2: the figure’s colour row only requires the four derived SDSS colour columns to be present in the final catalogue, giving 217 of 248 CG₄ galaxies, whereas the colour analysis also requires successful matched-photometry retention and the covariates used below. The fractions retained in this stricter colour-analysis subset are 44.4% in CG₄, 33.3% in Control_{4B}, 41.0% in Control_{4C}, 45.5% in RG₄.

The raw catalogue colour–mass fits indicate that CG₄ does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in all four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG₄ coefficients is significantly negative after Holm correction. Adding morphology and sSFR class does not reveal a significant negative coefficient. However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG–satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG₄ slope is not a robustly distinct population-level signature. Detailed colour–mass, satellite, and radial diagnostics are provided in the online supplement. Figure E.1 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG₄ satellites.

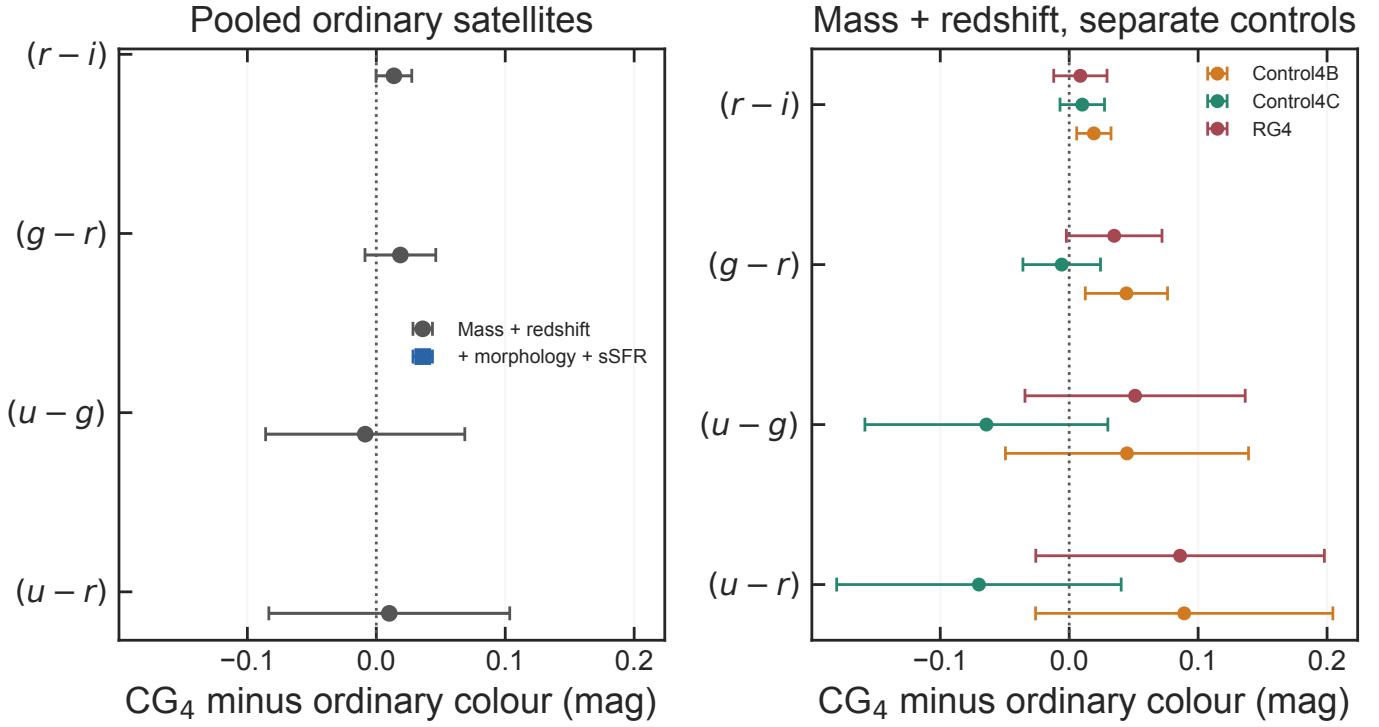


Fig. E.1. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG₄ minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG₄ satellites.

Appendix F: Pooled models, matching details, and selection diagnostics

F.1. Group-level multiplicity sensitivity

The three per-control group-level satellite-composition matches are treated in the main text as distinct control-definition estimands, and their p-values are not adjusted across Control_{4B}, Control_{4C}, and RG₄. Table F.1 reports the Holm sensitivity requested by that alternative convention. The pooled group-level match is excluded because it is a secondary summary of overlapping control catalogues.

F.2. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily (all RG₄ galaxies are also Control_{4B} members, and Control_{4B} and Control_{4C} share their cores), the pooled control side is first deduplicated by SDSS objid — one row per physical galaxy, unweighted, with every source label recorded — and standard errors are clustered by physical Lim group. The com-

mon adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion (all complete under the Sect. 2.4 rule). This pooled family is read as a secondary summary; the per-control contrasts of Sect. 3.2 are primary.

For quenched status, the CG₄ odds ratio is 1.35, with 95% confidence interval [0.98, 1.86] and Holm-corrected $p = 0.136$.

For quenched satellite status, the CG₄ odds ratio is 1.53, with 95% confidence interval [1.10, 2.11] and Holm-corrected $p = 0.043$.

For star-forming satellite status, the CG₄ odds ratio is 0.66, with 95% confidence interval [0.47, 0.91] and Holm-corrected $p = 0.043$.

For elliptical morphology, the CG₄ odds ratio is 1.77, with 95% confidence interval [1.32, 2.37] and Holm-corrected $p = 8.5 \times 10^{-4}$.

For elliptical satellite morphology, the CG₄ odds ratio is 2.08, with 95% confidence interval [1.51, 2.88] and Holm-corrected $p < 10^{-4}$.

For elliptical BGG morphology, the CG₄ odds ratio is 0.78, with 95% confidence interval [0.39, 1.53] and Holm-corrected $p = 0.464$.

For spiral morphology, the CG₄ odds ratio is 0.57, with 95% confidence interval [0.42, 0.76] and Holm-corrected $p = 8.5 \times 10^{-4}$.

For spiral satellite morphology, the CG₄ odds ratio is 0.48, with 95% confidence interval [0.35, 0.66] and Holm-corrected $p < 10^{-4}$.

Compact-group membership therefore remains independently associated with morphology after the available adjustments: CG₄ galaxies have a higher probability of being elliptical and a lower probability of being spiral. In contrast,

Table F.1. Holm sensitivity for the three per-control group-level satellite-composition matches. The adjusted values are computed from the full-precision bootstrap and paired sign-flip permutation p-values.

Control	p_{perm}	Holm	p_{perm}	p_{boot}	Holm	p_{boot}
Control _{4B}	0.016		0.047	0.012		0.036
Control _{4C}	0.080		0.080	0.082		0.082
RG ₄	0.026		0.053	0.019		0.038

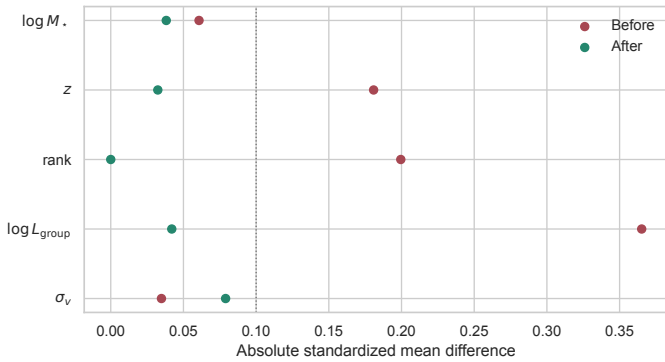


Fig. F.1. Absolute standardized mean differences before and after matching.

the adjusted quenched and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

F.3. Galaxy-level matching details

The galaxy-level control pool of Sect. 3.3 is deduplicated by SDSS objid before matching: 5828 control rows collapse to 3609 unique galaxies. The 14 unmatched CG₄ galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. By kept label, the matched controls comprise 159 Control_{4B}, 57 Control_{4C}, 18 RG₄; by physical provenance, 18 of the 234 matched controls are physically RG₄ galaxies (which also appear in Control_{4B} through the sample overlap), and the full per-galaxy provenance is recorded in the released `matched_control_provenance.csv`. The covariate balance before and after matching is shown in Fig. F.1; the per-outcome matched-effect figure is provided in the online supplement.

For completeness, the complement outcomes of the matched family are as follows.

The CG₄-minus-control difference in star-forming fraction is -0.031 , with 95% bootstrap interval $[-0.122, 0.056]$ and Holm-corrected $p = 1.000$.

The CG₄-minus-control difference in spiral fraction is -0.139 , with 95% bootstrap interval $[-0.240, -0.037]$ and Holm-corrected $p = 0.036$.

F.4. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Fig. F.2; this consolidated audit includes the Simard and Petrosian size availability used by Sect. 3.8. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG₄ and 255.9 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 7.7%, respectively.

Under the availability definitions shown in Fig. F.2, no major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity

dispersion, group luminosity, and BGG luminosity dominance. The tables do carry an M_{200c} estimate, but it is the Yang-Lim abundance-matching M_{180m} of the identified *host* halo (converted as in Paper I, Appendix A), not a quartet property: for embedded quartets it describes the richer host while for isolated systems it describes the quartet itself, it derives from group luminosity by abundance matching and is therefore nearly collinear with the group-luminosity covariate already in the models, and it is entangled with the compact/loose selection. We therefore exclude it from the harmonised covariate set. A sensitivity fit adding $\log M_{200c}$ to the published covariates leaves every elliptical conclusion unchanged: the pooled odds ratio moves from 1.77 to 1.62 ($p = 0.0020$), and the per-control odds ratios become 2.00, 1.56, and 2.65 against Control_{4B}, Control_{4C}, and RG₄ (all $p < 0.005$).

Restricting the pairwise tidal index of Sect. 3.6 to quartet co-members is likewise innocuous: recomputing T_i with all spectroscopic Lim members of the identified host group included (positions from the group catalogue; stellar masses from MPA-JHU where available, so the host-inclusive index is itself a lower bound) shifts $\log T_i$ by a median of 0.009 dex (sample medians ≤ 0.02 dex, and identically zero for RG₄, whose groups have exactly four members), with rank correlation 0.91 between the two definitions — the residual reordering is confined to a minority of loose quartets embedded in rich hosts — and refitting the with-index elliptical model on the host-inclusive index moves the attenuated CG₄ coefficient only from 1.39 to 1.33. T_i should thus be read as the local, galaxy-scale tidal density traced by catalogued galaxies; any smooth host-halo contribution is not measured by this sum and is probed separately through the host-mass sensitivity fit above and the within-host analysis of Sect. 3.4.

The colour-selection mass audit is provided in the online supplement.

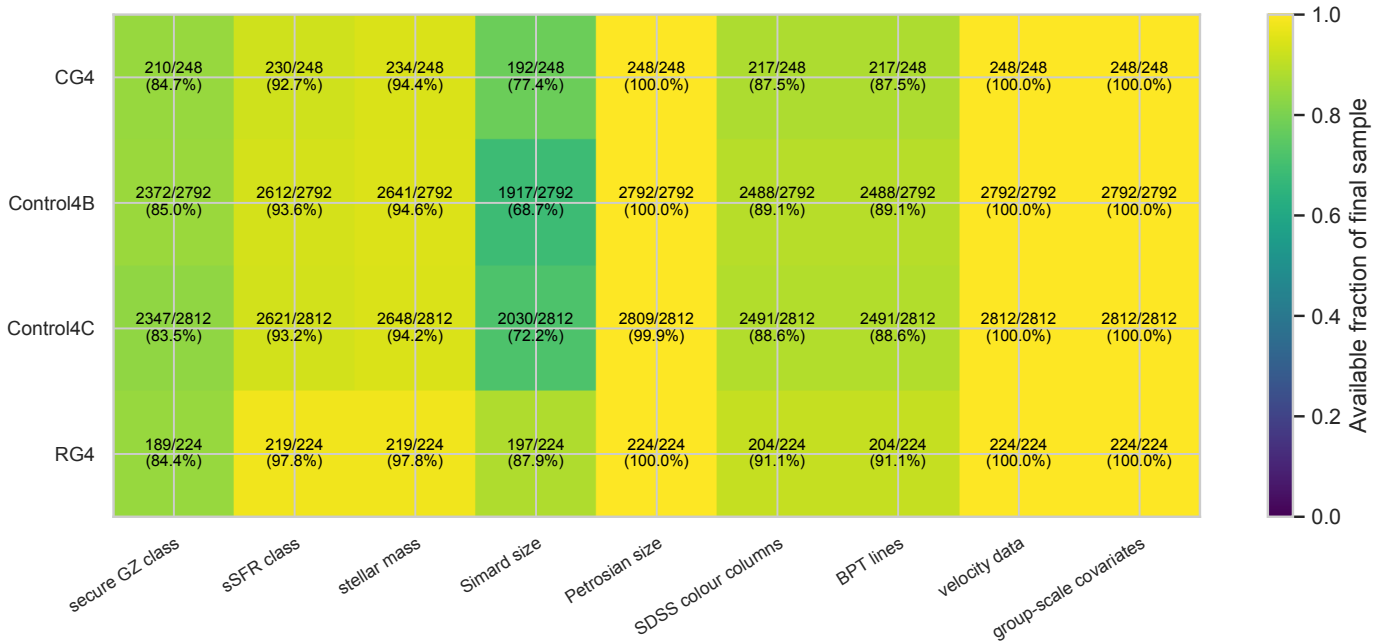


Fig. F.2. Availability fractions for the major derived quantities in each sample, including the Simard and Petrosian size-availability columns used by the size analysis. Cell labels give the number available over the final per-sample galaxy count and the corresponding percentage. The colour row records completeness of the four derived SDSS colour columns in the final catalogue, rather than the stricter colour-analysis matched subset used in Appendix E.6. The Simard-size column reflects the DR7 bridge, redshift, quality-window, blending, and pegged-Sérsic-index cuts described in Sect. 2.3.

Appendix G: Extended size diagnostics

The size-availability counts are included in the consolidated audit of Fig. F.2. This appendix therefore focuses on the extended size-effect diagnostics and on the contextual R_e –Sérsic- n plane; the descriptive mass–size relations and the per-sample sizes predicted at $\log(M_\star/M_\odot) = 10.3$ are provided in the online supplement.

The crowding checks mirror Sect. 3.5: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of -0.013 dex ($p = 0.463$), and adding the close-neighbour flag as a covariate gives -0.011 dex ($p = 0.479$).

The per-galaxy difference between the two measures, $d = \log R_{\text{chl},r} - \log R_{50,r}$, averages 0.054 dex, and is larger for elliptical galaxies (0.074 dex) than for spiral galaxies (0.042 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_s = 0.012$ ($p = 0.524$), so residual blending does not drive a strong measure-dependent size systematic.

Adding the pairwise projected tidal index of Sect. 3.6 to the all-galaxy size model changes the CG_4 coefficient from -0.012 dex to 0.005 dex on the pairwise complete-case sample; the local projected-density proxy attenuates the size coefficient in the same way it attenuates the morphology coefficient.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_s = 0.17$ ($p = 0.026$) in CG_4 and $\rho_s = 0.18$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4 -specific inner-region size suppression.

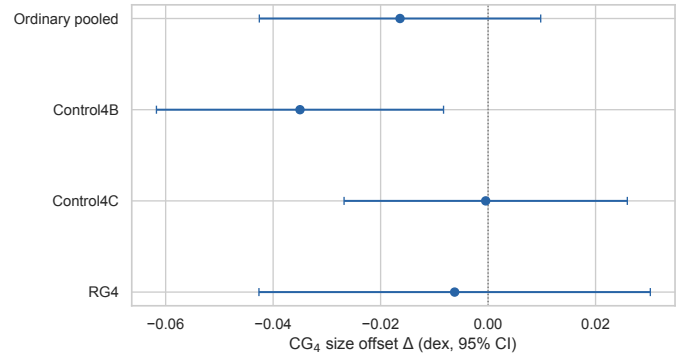


Fig. G.1. Mass- and redshift-adjusted CG_4 size coefficients against the pooled controls and against each control catalogue separately. Bars show 95% confidence intervals; negative values mean smaller CG_4 sizes at fixed mass.

Within morphology classes, the satellite-only fits give 0.036 dex for elliptical satellites (Holm-corrected $p = 0.205$) and -0.005 dex for spiral satellites (Holm-corrected $p = 0.831$), and the $\text{CG}_4 \times \text{morphology}$ interaction is 0.032 dex ($p = 0.348$). These strata are secondary diagnostics; they test whether any size offset is independent information beyond the morphology signal rather than a restatement of it.

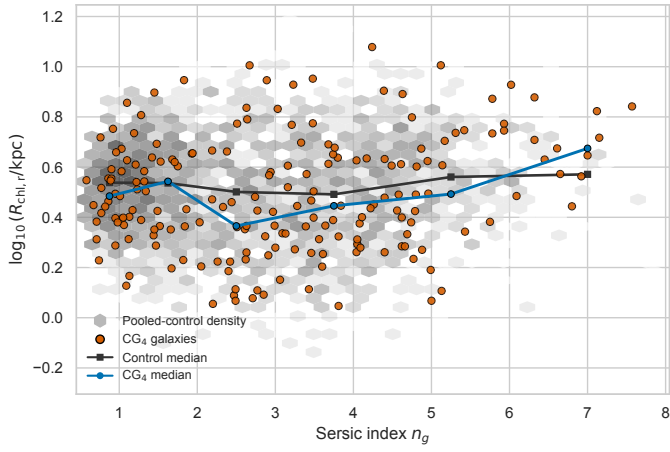


Fig. G.2. Half-light radius versus Sérsic index for CG₄ and the pooled controls, shown for galaxies with both valid Simard size and stellar mass in the mass-size analysis. The pooled controls are shown as a grey logarithmic hexbin density, while CG₄ galaxies are overplotted as foreground points; lines show binned medians where bins are sufficiently populated. This is the plane used by the S-PLUS transition-galaxy studies.